

Evaluating AI Guardrails and Their Effect on Students' Independent Critical Thinking

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Abstract— Artificial intelligence tools such as ChatGPT are increasingly used by students for problem solving and learning. While these tools improve efficiency, they raise concerns about reduced independent thinking and over-reliance on generated answers. This study investigates whether structured AI guardrails can help maintain critical thinking in AI-assisted learning. A three-group experimental design with college students compares unrestricted AI use, AI use with guardrails, and no AI usage. The guardrails focus on verification, explanation, and reflection. Preliminary results indicate that students using guardrails verify AI outputs more frequently and provide stronger reasoning compared to unrestricted AI use. An additional informal observation with a young learner shows similar behavioral patterns. These findings suggest that structured guidance can help balance the benefits of AI while preserving independent thinking.

1 INTRODUCTION

AI has quickly become part of everyday learning. Students use tools like ChatGPT to solve problems, summarize information, and understand new concepts. While this can be helpful, it also raises a key concern. When students receive answers instantly, they may not fully engage with the reasoning behind those answers.

From my own experience conducting technical workshops and learning sessions, I have observed a consistent pattern. When AI is available without any structure, students tend to accept outputs quickly. However, when they are asked to explain or justify those outputs, their behavior changes. They slow down, verify results, and think more carefully.

This led to a central question:

Can we guide how students use AI so that it supports thinking instead of replacing it?

To address this, this work introduces AI guardrails, a simple structured approach that encourages students to:

- verify AI-generated outputs
- explain reasoning in their own words
- reflect on how AI was used

The goal is not to restrict AI, but to use it in a way that keeps students actively involved in the learning process. This work focuses not only on whether AI affects learning, but on how student interaction with AI can be designed to preserve critical thinking.

2 RELATED WORK

Recent studies have examined the growing influence of artificial intelligence on student learning and critical thinking. Systematic reviews such as [1] highlight concerns about over-reliance on AI and its potential to reduce independent reasoning in higher education settings. Empirical work like [2] further explores how AI tools affect students' thinking processes and learning behaviors in practice.

At the same time, several researchers have proposed approaches to guide the use of AI in educational settings. For instance, [4] discusses the transformative potential of AI in enhancing learning while raising concerns about its impact on critical thinking. Similarly, [3] introduces a pedagogical framework aimed at safeguarding cognitive skills, emphasizing the importance of structured interaction with AI. More directly, [5] explores the design of safety guardrails in AI-based educational tools, highlighting the need for mechanisms that support responsible AI usage.

However, much of the existing work focuses on conceptual frameworks or system design without experimentally validating how such guardrails influence student behavior and learning outcomes. In contrast, this study introduces a

guardrail-based approach that guides student interaction with AI through verification, explanation, and reflection, and evaluates its effectiveness using a controlled experimental design.

By focusing on measurable behaviors such as verification frequency and reasoning quality, this work moves beyond conceptual discussion and provides an actionable and testable approach to integrating AI in education.

Related Work References	Strength	Gap addressed by this paper
[1] Critical Thinking in the Age of AI: A Systematic Review of AI's Effects on Higher Education. Educational	Provides a comprehensive overview of AI's impact on critical thinking through a systematic review	Does not include experimental validation or propose a concrete intervention
[2] The Influence of Artificial Intelligence on Students' Critical Thinking	Examines how AI influences student thinking through an empirical study	Lacks structured guidance or a framework to improve student interaction with AI
[3] Critical Alliance of AI in Education: A Pedagogical Framework for Safeguarding Cognitive Skill	Proposes a pedagogical framework to safeguard cognitive skills in AI-supported learning	Conceptual in nature; does not evaluate the framework through controlled experiments
[4] Advancing Critical Thinking in Higher Education: The Transformative Role of Artificial Intelligence	Highlights both opportunities and risks of AI in enhancing learning and critical thinking	Focuses on discussion and perspective without testing practical solutions
[5] Building effective safety guardrails in AI education tools	Introduces the concept of safety guardrails in AI educational tools	Focuses on system design; does not measure impact on student reasoning or learning outcomes

3. METHODOLOGY

3.1 Study Design

The primary study was conducted with college students participating in learning sessions. A total of 21 participants were included in the study, with approximately equal distribution across the three experimental groups.

Participants were divided into three groups:

Group	Condition
A	Unrestricted AI use
B	AI use with guardrails
C	No AI use

Each participant completed:

1. Pre-test without AI
2. Learning phase based on assigned condition
3. Post-test without AI
4. Survey and reflection

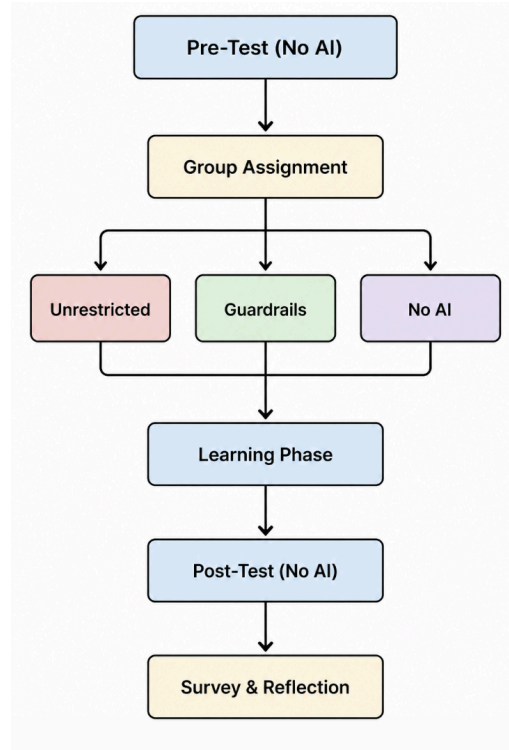


Figure 1— Figure 1. Experimental workflow illustrating participant flow across study conditions

3.2 Guardrail Framework

The guardrails introduce a structured interaction pattern:

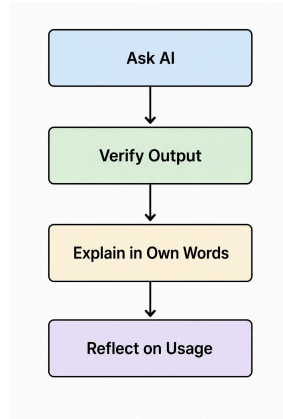


Figure 2— AI use with guardrails architecture

3.3 Evaluation Metrics

Metric	Description
Post-test performance	Independent problem-solving ability
Verification frequency	How often students check AI outputs
Explanation quality	Depth and clarity of reasoning
AI reliance	Self-reported dependency

3.4 Learning Phase

The study was conducted with college students who were actively learning machine learning concepts through coursework and technical training sessions. The learning phase involved tasks aligned with typical machine learning topics, allowing students to apply conceptual understanding, interpret results, and explain their reasoning.

3.5 Post AI Sample questions

Students were tested on 5 Machine learning based questions. The same level of difficulty and conceptual focus was maintained across all groups to ensure fair comparison. Below are 3 of sample questions they were asked:

Sample Question 1: A classification model achieves 95% accuracy on training data but 60% accuracy on test data. Explain what is happening and suggest two ways to improve the model.

Sample Question 2: You are given a confusion matrix for a binary classifier. Explain what precision and recall represent and describe a scenario where recall is more important than precision.

Sample Question 3: When would you prefer a decision tree over a neural network? Explain your reasoning.

These questions tested students' understanding of overfitting, reasoning, solution thinking, conceptual clarity, and real-world reasoning.

4 RESULTS

The results presented are based on a pilot study and are intended to highlight early trends rather than definitive conclusions.

4.1 Behavioral Trends

From the pilot study:

- Students in the guardrail group verified outputs more frequently
- Students in the unrestricted group relied heavily on AI
- Students without AI worked more independently but took longer

4.2 Explanation Quality, Verification Behavior and Score

Group	How well a student can explain the answer in their own words	How often and how actively a student checks whether the AI answer is correct	Median Post Study Test Scores without AI (out of 10)
Unrestricted AI	Often repeated AI reasoning	Rarely	5
Guardrails	Provided clearer explanations	Always	8
No AI	Independent but sometimes less accurate	N/A	6

4.3 Survey and Reflection Insights

Survey and reflection responses provided additional insight into how students perceive and interact with AI tools.

Most participants reported frequent use of AI tools for tasks such as understanding concepts, summarizing content, and solving problems. While a majority of students indicated that they attempt to verify AI-generated responses, the depth of verification varied across individuals.

Students in the guardrail group reported a higher tendency to actively question AI outputs and reflect on their reasoning process. Many participants noted that the requirement to explain answers in their own words helped them better understand the underlying concepts rather than simply accepting the result.

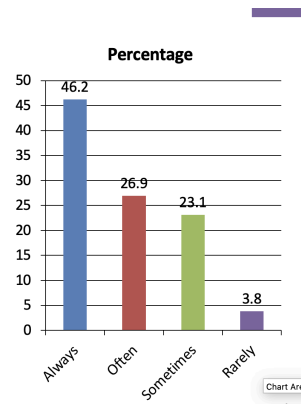
In contrast, students in the unrestricted AI group often reported relying on AI for faster task completion, with less emphasis on validating correctness. Although

they expressed confidence in their answers, this confidence did not always align with their explanation quality observed in the post-test.

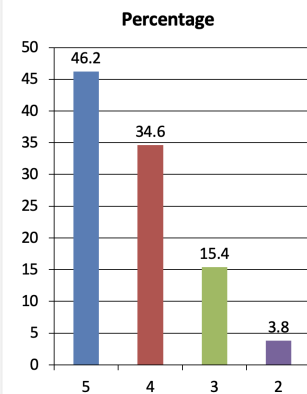
An interesting observation across participants was a strong agreement that AI should be used with structured guidelines rather than freely. Many students also indicated that training on how to verify AI outputs would improve their learning experience.

These findings suggest that while students are aware of the need to verify AI outputs, structured guidance plays a key role in translating that awareness into consistent behavior.

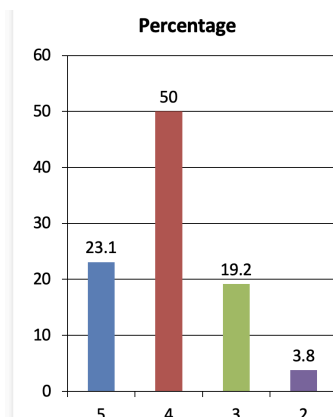
How often do you verify AI-generated answers?



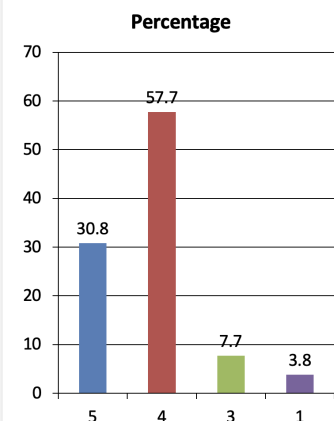
I try to understand the reasoning behind AI-generated answers.



I feel confident solving similar problems without AI.



Verification training would improve my learning.



5 DISCUSSION

The results suggest that how AI is used is more important than whether it is used.

Unrestricted AI allows students to complete tasks quickly, but often reduces engagement with the reasoning process. In contrast, not using AI promotes independence but can slow down learning and reduce efficiency.

The guardrail approach provides a balance. It allows students to use AI while still encouraging them to think, verify, and explain. This leads to deeper engagement without removing the benefits of AI.

Another observation is that students often feel confident in their understanding, but this confidence does not always reflect actual reasoning ability. Guardrails help address this gap by requiring interaction with the solution process.

An important observation is that increased verification behavior was associated with higher explanation quality, suggesting that students who actively questioned AI outputs developed deeper understanding.

6 EXPLORATORY OBSERVATION: EARLY LEARNER CASE STUDY

In addition to the college study, an informal observation was conducted in a home setting with an 8-year-old learner to explore whether similar patterns appear at an earlier stage.

6.1 Setup

- Task: simple multiplication problems
- AI access: allowed using ChatGPT

6.2 Phase 1: Without Guardrails

The learner was allowed to freely use AI. Some of the observations made are : student relied on AI for answers, limited effort to verify, treated AI output as final

6.3 Phase 2: With Guardrails

A structured approach was introduced to check correctness, explain reasoning and attempt independently. With this model student was more engagement with the problem, asked follow-up questions and attempted to understand the steps

Condition	Behavior	Score
No Guardrails	Fast but dependent	6/10
With Guardrails	Slower but more engaged	9/10

6.4 Interpretation

This observation is not intended as formal evidence, but it highlights a similar pattern to the college study. Even at an early learning stage, structured guidance appears to influence how learners interact with AI.

7 LIMITATIONS

- Small sample size in the pilot study
- Short duration of intervention
- Informal nature of the exploratory observation

8 FUTURE WORK

Future work will focus on:

- Expanding the study to a larger population
- Conducting full statistical analysis
- Testing across different age groups
- Designing scalable AI-based learning assistants with built-in guardrails
- Due to the pilot nature of the study and small sample size, the analysis focuses on descriptive statistics. Future work will include statistical validation using methods such as ANOVA.

9 CONCLUSION

AI is becoming an integral part of education. The key challenge is not whether to use it, but how to use it effectively.

This study shows that simple guardrails can encourage students to verify, explain, and think more deeply. Instead of reducing learning, AI can support it when used with the right structure.

The consistency between the college study and the early learner observation suggests that this pattern may extend across different learning stages. With thoughtful design, AI can become a tool that strengthens critical thinking rather than replacing it.

10 REFERENCES

- [1] Melisa, R., Ashadi, A., Triastuti, A., Hidayati, S., Salido, A., Ero, P. E. L., ... & Al Fuad, Z. (2025). Critical Thinking in the Age of AI: A Systematic Review of AI's Effects on Higher Education. *Educational Process: International Journal*, 14, e2025031
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